



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (semester I)

Class: F.Y B. Tech

Course: Process Equipment Design &
Drawing-II

Course Code: PEC-CH401

Date: 14.10.2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.

Question		Marks	CO	BT
Q.1	I. Define material handling? II. What are the objectives of material handling III. Explain the principle of material handling. IV. Summarize unit load concept.	08	CO2	BT2
	OR I. What is plate heat exchanger II. Illustrate plate heat exchanger parts and its functions III. Which benefits offered by plate heat exchanger			
Q.2	In shell and tube heat exchanger hot lubricating oil cooled from $140^{\circ}\text{C} \rightarrow 85^{\circ}\text{C}$ (Hot fluid inlet/outlet (lubricating oil): $T_{h,in} = 140^{\circ}\text{C}$, $T_{h,out} = 85^{\circ}\text{C}$) by cold seawater $30^{\circ}\text{C} \rightarrow 65^{\circ}\text{C}$ (Cold fluid inlet/outlet (seawater): $T_{c,in} = 30^{\circ}\text{C}$, $T_{c,out} = 65^{\circ}\text{C}$). Hot lubricating oil mass flow rate is 140,000 kg/h. Specific heats (assumed): $C_{p,h} = 2200 \text{ J/kg.K}$ (oils), $C_{p,c} = 4180 \text{ J/kg.K}$ (water). Heat exchanger Tube geometry (assumed): tube outer diameter $D_o = 0.025 \text{ m}$, tube inner diameter $d_i = 0.021 \text{ m}$, tube length $L = 5.00 \text{ m}$, triangular pitch $1.25 D_o$. Configuration is: 1 shell pass, 2 tube passes (1S-2T). Trial overall heat-transfer coefficient $U_{trial} = 420 \text{ W/m}^2\text{K}$, Correction factor for 1S-2T (from Kern charts) $F = 0.85$ and $K_1 = 0.249$, $n_1 = 2.207$. Clearance is 68mm. So, estimate tube-bundle diameter to design shell and tube heat exchanger by Kern method.	07	CO3	BT5
Q.3	Define following parts of shell and tube heat exchanger I. Shell, Tubes, Tube sheets, Baffles, Tube pitch, Tube clearance, Expansion joint	05	CO2	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the principles and theories combined with a practical knowledge of the limits imposed by environmental, safety and health concerns to design of new process and expansion and revision of the existing process
CO 2	Conveniently use of various codes and standards in design and their application in designing new processes.
CO 3	Able to design various process equipment's and their requisite accessories as per standards.
CO 4	Able to simulate the process parameters